MATH 111 Calculus 1, NUU, FALL 2026

Dr. Salnikova Chekeres **Tutorial 1**

Revision of Chapter 1, Sections 1.1–1.3 (Functions and models) and Chapter 1 Review: Concept Check 1–11, Exercises 1–24. Two sessions of 50 minutes.

Key facts to have at hand

- A **function** f assigns to each element x of a set D (the **domain**) exactly one value $f(x)$. The **range** is $\{f(x) : x \in D\}$. A curve is the graph of a function if and only if no vertical line meets it more than once (**vertical line test**).
- Four ways to represent a function: **verbally**, **numerically** (table), **visually** (graph), **algebraically** (formula).
- f is **even** if $f(-x) = f(x)$ (graph symmetric about the y-axis) and **odd** if $f(-x) = -f(x)$ (symmetric about the origin). f is **increasing** on I if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$ in I.
- Catalogue of models: linear $mx + b$, polynomial, power x^a, rational $P(x)/Q(x)$, algebraic, trigonometric, exponential b^x (logarithmic and inverse functions come later in the course).
- Transformations of $y = f(x)$, with $c > 0$: $f(x) + c$ up, $f(x) - c$ down, $f(x - c)$ right, $f(x + c)$ left; $cf(x)$ stretch vertically ($c > 1$) or shrink ($0 < c < 1$); $f(cx)$ shrink horizontally ($c > 1$) or stretch ($0 < c < 1$); $-f(x)$ reflect about the x-axis; $f(-x)$ reflect about the y-axis.
- Combinations: $(f \pm g)(x) = f(x) \pm g(x)$, $(fg)(x) = f(x)g(x)$ on $\mathrm{dom}f \cap \mathrm{dom}g$; $(f/g)(x) = f(x)/g(x)$ where additionally $g(x) \neq 0$. Composition: $(f \circ g)(x) = f(g(x))$, whose domain is $\{x \in \mathrm{dom}g : g(x) \in \mathrm{dom}f\}$.

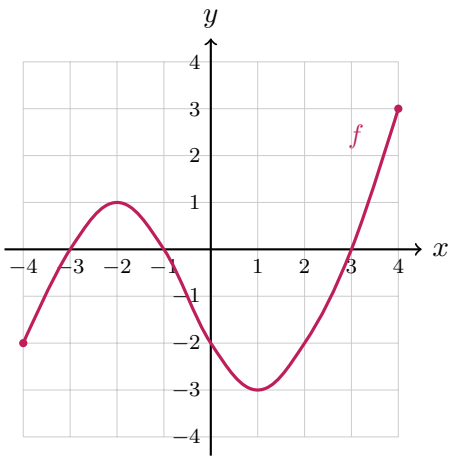
Part I · Functions and models (Sections 1.1–1.2) · 50 minutes

Warm-up I · True or false? (justify with one sentence or one counterexample)

1. If f is a function, then $f(2x) = 2f(x)$ for all x.
2. If $f(a) = f(b)$, then $a = b$.
3. A horizontal line can cross the graph of a function more than once.
4. Every curve that passes the vertical line test is the graph of a function.
5. If a function is both even and odd, then $f(x) = 0$ for every x in its domain.
6. The domain of $\sqrt{x^2 - 1}$ is $[1, \infty)$.
7. The function $f(x) = 1/x$ is decreasing on its whole domain.
8. The graph of $y = x^4$ lies above the graph of $y = x^2$ for every $x \neq 0$.
9. $f(x) = 2 \cdot 3^x$ is a power function.
10. The range of $f(x) = 3 + 2\sin x$ is $[1, 5]$.

Problem 1 *Reading a graph.*

The graph of a function f is shown.



(a) Find $f(-2)$, $f(1)$ and $f(4)$.

(b) Find all x with $f(x) = 0$. Then estimate all x with $f(x) = 1$.

(c) State the domain and the range of f in interval notation.

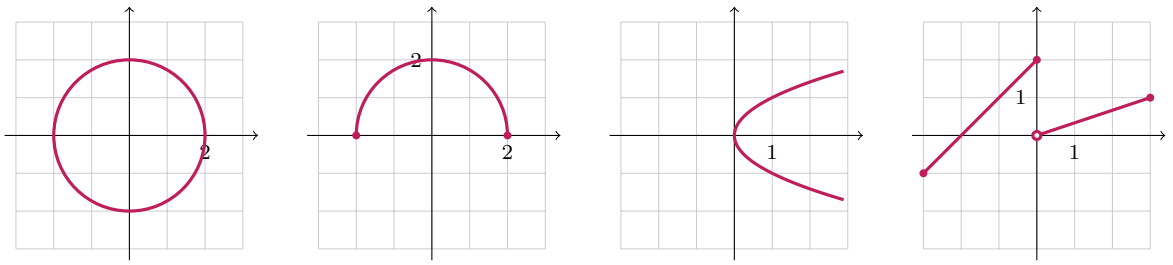
(d) On which intervals is f increasing? On which is it decreasing?

(e) Is f even, odd, or neither? Explain.

(f) How many solutions does the equation $f(x) = -2$ have? Which are they?

Problem 2 *Is it the graph of a function?*

For each curve decide whether it is the graph of a function of x. If it is, state the domain and the range.



(i) (ii) (iii) (iv)

Problem 3 *Difference quotients.*

Let $f(x) = 2x^2 - 3x + 1$.

(a) Compute and simplify $\dfrac{f(a+h) - f(a)}{h}$ $(h \neq 0)$.

(b) Compute and simplify $\dfrac{f(x) - f(2)}{x - 2}$ $(x \neq 2)$.

(c) What value does the quotient in (a) approach as h gets closer and closer to 0? What do you think this number describes about the graph of f at $x = a$?

Problem 4 *Domain and range.*

Find the domain of each function; write the answer in interval notation.

(a) $f(x) = \dfrac{x+1}{x^2 - 5x + 6}$

(b) $g(x) = \sqrt{2x - 6}$

(c) $h(x) = \dfrac{\sqrt{9 - x^2}}{x + 1}$

(d) $k(t) = \dfrac{1}{\sqrt{t^2 - 4}}$

(e) $m(x) = \sqrt{x} + \sqrt{3 - x}$

Find the domain *and* the range of

(f) $F(x) = \dfrac{3}{2x - 5}$

(g) $G(x) = \sqrt{25 - x^2}$

(h) $H(t) = 2 - 3\sin 2t$

Problem 5 *Piecewise-defined functions.*

(a) Sketch the graph of

$$f(x) = \begin{cases} x + 3, & x < -1, \\ x^2, & -1 \le x \le 1, \\ 2 - x, & x > 1, \end{cases}$$

and find $f(-1)$, $f(1)$, $f(3)$ and the range of f.

(b) Write $g(x) = |2x - 4|$ as a piecewise-defined function without absolute values and sketch its graph.

(c) A car park charges 5,000 UZS for any stay of up to one hour and 3,000 UZS for each additional hour or part of an hour, for stays of up to five hours. Sketch the cost $C(t)$ of a stay of t hours, $0 < t \le 5$, and write a formula for $C(t)$.

Problem 6 *Building a function from a description.*

In each case express the required quantity as a function of the given variable and state the domain that makes sense in the situation.

(a) A rectangle has perimeter 24 m. Express its area A as a function of the length x of one of its sides.

(b) An open box is made from a 30 cm × 30 cm square sheet of cardboard by cutting equal squares of side x from the four corners and folding up the sides. Express the volume V of the box as a function of x.

(c) A closed cylindrical can must hold 500 cm^3. Express the total surface area S (top, bottom and side) as a function of the radius r.

(d) Express the area of an equilateral triangle as a function of the length s of a side.

Problem 7 *Mathematical models.*

(a) A bakery finds that it costs 2,400,000 UZS to produce 300 loaves of bread in a day and 3,000,000 UZS to produce 500 loaves. Assume that the daily cost C is a linear function of the number x of loaves.

(i) Find $C(x)$ and sketch its graph.

(ii) What is the slope of the graph and what does it represent?

(iii) What is the C-intercept and what does it represent?

(b) On a summer afternoon the air temperature at ground level in Tashkent is 36 °C, and in the lower atmosphere the temperature drops by about 6.5 °C for every kilometre of altitude. Express the temperature T as a function of the altitude h (in km, $0 \le h \le 11$). At what altitude is the temperature 0 °C?

(c) Classify each function as linear, polynomial (state the degree), power, rational, algebraic, trigonometric or exponential:

$$f(x) = x^5 - 2x + 7, \qquad g(x) = x^{-2/3}, \qquad h(x) = \frac{x^2+1}{x^3 - x}, \qquad k(x) = 5 \cdot 2^x,$$
$$m(x) = \tan x - 1, \qquad n(x) = x\sqrt{x^2+1}, \qquad p(x) = 3x - 4.$$

Which of x^π and π^x is a power function?

(d) The number of daylight hours in Tashkent on day t of the year ($t = 1$ is January 1) is modelled by

$$D(t) = 12.2 + 2.8\sin\Big(\frac{2\pi}{365}(t - 80)\Big).$$

(i) What are the amplitude and the period of D? What is the longest and the shortest day according to the model, and approximately on which dates do they occur?

(ii) For which values of t does the model give exactly 12.2 hours of daylight? What are these days called?

Part II · New functions from old (Section 1.3) · 50 minutes

Warm-up II · True or false, then track the point

1. The graph of $y = f(x - 3)$ is the graph of f shifted 3 units to the left.
2. The graph of $y = f(2x)$ is obtained from the graph of f by stretching it horizontally by a factor of 2.
3. For all functions f and g, $f \circ g = g \circ f$.
4. If f is even, then $y = f(x - 1)$ is also even.
5. If f and g are both odd, then $f \circ g$ is odd.
6. If f and g are both odd, then the product fg is even.
7. The domain of $f \circ g$ is the same as the domain of g.
8. $\sqrt{x^2} = x$ for every real number x.

Track the point. The point $(1, 2)$ lies on the graph of $y = f(x)$. Find the corresponding point on the graph of

(i) $y = f(x) + 3$, (ii) $y = f(x + 3)$, (iii) $y = 2f(x)$, (iv) $y = f(2x)$,
(v) $y = -f(x)$, (vi) $y = f(-x)$, (vii) $y = f\big(\frac{x}{2}\big) - 1$, (viii) $y = 3f(x - 1)$.

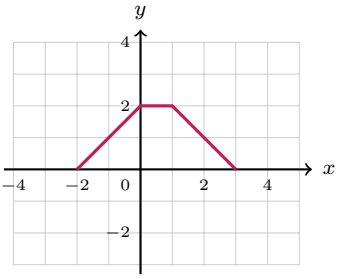
Problem 8 *Describe the transformation.*

Suppose the graph of f is given. Describe in words how the graph of each function is obtained from it. Where two steps are needed, give them in a correct order.

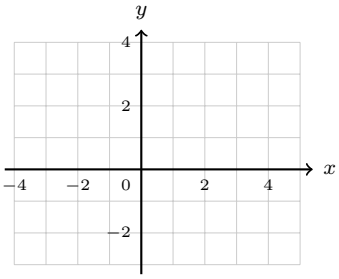
(a) $y = f(x) - 4$
(b) $y = f(x + 4)$
(c) $y = 3f(x) + 1$
(d) $y = f(x - 1) - 2$
(e) $y = -f(x)$
(f) $y = f(-x)$
(g) $y = f(3x)$
(h) $y = 1 - f(x)$
(i) $y = |f(x)|$

Problem 9 *Draw the transformed graphs.*

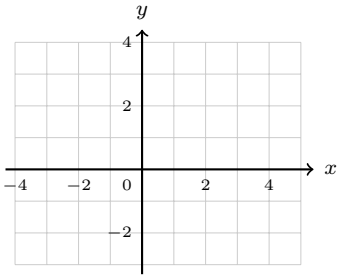
The graph of f consists of the segments joining $(-2,0)$, $(0,2)$, $(1,2)$ and $(3,0)$, as shown in the first picture. Draw the graph of each function on its own grid.



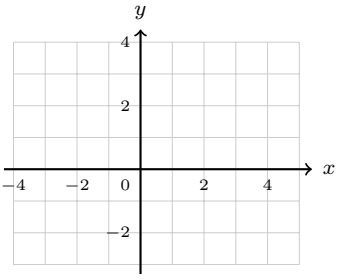
$y = f(x)$



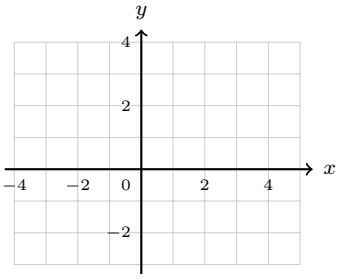
(a) $y = f(x) + 1$



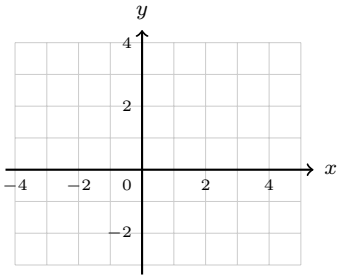
(b) $y = f(x - 2)$



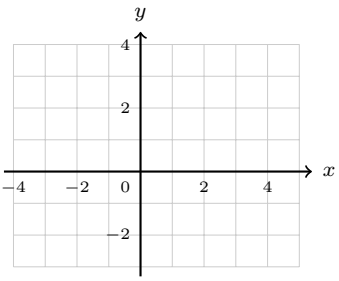
(c) $y = -f(x)$



(d) $y = f(2x)$



(e) $y = \frac{1}{2}f(x) - 1$



(f) $y = f(-x)$

In each case first write down where the four corner points go; then join them. Check: which transformations change the domain $[-2, 3]$, and to what?

Problem 10 *Sketch with transformations.*

Name the basic function you start from, list the transformations in order, and sketch the graph. Mark the key points (vertex, intercepts, asymptotes where relevant).

(a) $y = (x+1)^3 - 2$

(b) $y = 2\sqrt{x-3}$

(c) $y = x^2 + 4x + 1$

(d) $y = 1 - \sin 2x$

(e) $y = \dfrac{1}{x-2} + 1$

(f) $y = 2 - |x+1|$

(g) $y = \left|x^2 - 4\right|$

Problem 11 *Even, odd or neither?*

Decide algebraically, and say what the answer means for the graph.

(a) $f(x) = x^4 - 3x^2 + 1$

(b) $g(x) = x^3 + x$

(c) $h(x) = x^2 + x$

(d) $k(x) = x\cos x$

(e) $m(x) = |x| + 1$

(f) $p(x) = 1 + \sin x$

(g) $q(x) = \dfrac{\sin x}{x}$

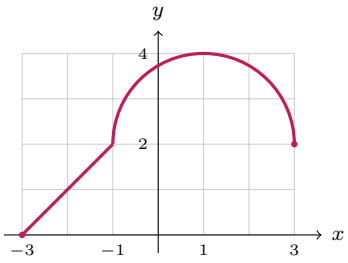
(h) $r(x) = \dfrac{x}{x^2+1}$

Then prove: if f and g are both odd, then $f \circ g$ is odd and fg is even. What can you say about $f \circ g$ if f is even (and g is any function)?

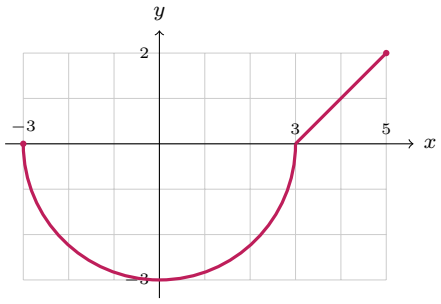
Problem 12 *A formula from a picture.*

(a) Find an expression for the function whose graph consists of the line segment from $(-3, 0)$ to $(-1, 2)$ together with the top half of the circle with centre $(1, 2)$ and radius 2.

(b) Find an expression for the function whose graph consists of the bottom half of the circle $x^2+y^2 = 9$ together with the line segment from $(3, 0)$ to $(5, 2)$.



(a)



(b)

Problem 13 *Combining functions.*

Let $f(x) = \sqrt{4-x}$ and $g(x) = \sqrt{x+1}$. Find $f+g$, $f-g$, fg and f/g, with their domains.

Problem 14 *Composition.*

(a) Let $f(x) = \sqrt{x-1}$ and $g(x) = x^2+1$. Find $f \circ g$, $g \circ f$, $f \circ f$ and $g \circ g$ and their domains. Simplify where possible — and be careful: does simplifying change the domain?

(b) Let $h(x) = \sin x$ and $k(x) = \dfrac{1}{x}$. Find $h \circ k$ and $k \circ h$ and their domains.

Problem 15 *Decomposing functions.*

(a) Write $F(x) = (1+\sqrt{x})^5$ as $f \circ g$, and then as $f \circ g \circ h$ with three "simple" functions.

(b) Write $G(x) = \sin^2(3x+1)$ as $f \circ g \circ h$.

(c) Write $H(x) = \dfrac{1}{\sqrt{x^2+4}}$ as $f \circ g \circ h$.

(d) If $f(x) = x^2$ and $(f \circ g)(x) = 4x^2 - 12x + 9$, find a possible g. Is it the only one?

Problem 16 *Transformations as a recipe.*

(a) Start with $y = x^2$. Reflect about the x-axis, stretch vertically by a factor of 3, shift 2 units to the left and 1 unit up. Write the equation of the resulting graph and give its vertex.

(b) Does the order matter? Apply to $y = x^2$ first "shift 1 up, then stretch vertically by 3", and then the same two steps in the opposite order. Compare the results.

(c) Find a function whose graph is the graph of $y = 1/x$ shifted so that its asymptotes are the lines $x = 1$ and $y = -2$.

Challenge *(for groups that finish early, or at home)*

(a) Let $f(x) = 2x+1$. Compute $f \circ f$ and $f \circ f \circ f$. Guess a formula for the n-fold composition $f_n = f \circ f \circ \cdots \circ f$ (n copies of f) and check it for $n = 4$.

(b) Sketch the graph of $g(x) = |x| - |x - 2|$ by writing it as a piecewise-defined function.

After the tutorial. Stewart, Chapter 1 Review: Concept Check 1–11 (say each answer aloud to a classmate), the True–False Quiz items on functions, and Exercises 1–24. The problems on this sheet are modelled on those exercises, so if you can do the sheet you can do them — and the other way round.